

Product Carbon Footprint Study Report

Product Name: fan, for power supply unit, desktop computer

Producer Name: [Information provided by the user]

Report No: [Information provided by the user]

Date: September 04, 2026

I. General Information

Producer Name: [Information provided by the user]

Address: [Information provided by the user]

Legal Representative: [Information provided by the user]

Product Name: fan, for power supply unit, desktop computer

Product Function: [Information provided by the user]

Based on the Standard: ecoinvent - EF v3.1

II. Quantitative Purpose

1kg fan, for power supply unit, desktop computer

III. Scope of Quantification

1. Functional or Declaratory Units

Functional or declarative units in 1kg fan, for power supply unit, desktop computer.

2. System Boundary

fan production, for power supply unit, desktop computer, in China in 2025

3. Cut-off Criteria

[Information provided by the user]

4. Time Scale

[Information provided by the user]

IV. Inventory Analysis

1. Description of Data Sources

Primary Data: [Information provided by the user]

Secondary Data: ecoinvent

2. Principles and Procedures for Allocation

Basis of Allocation: [Information provided by the user]

Allocation Procedure: [Information provided by the user]

Specific Allocations: [Information provided by the user]

3. Data Quality Evaluation (Optional)

[Information provided by the user]

V. Impact Assessment

1. Impact Type and Characterization Factor Selection

100-year global warming potential (GWP) given by IPCC

2. Product Carbon Footprint Results Calculation

The life cycle carbon footprint of fan, for power supply unit, desktop computer produced by [Information provided by the user], per functional unit of product, from raw material acquisition to end of life is **8.287 kg CO2-Eq**.

Figure 1: Carbon Emissions Distribution by Life Cycle Stage

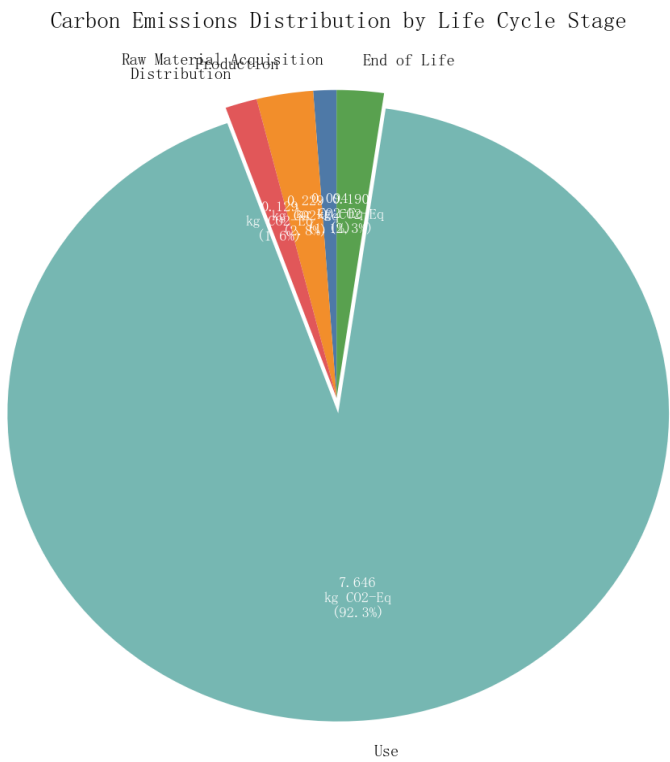


Table 1: Carbon Emissions by Life Cycle Stage

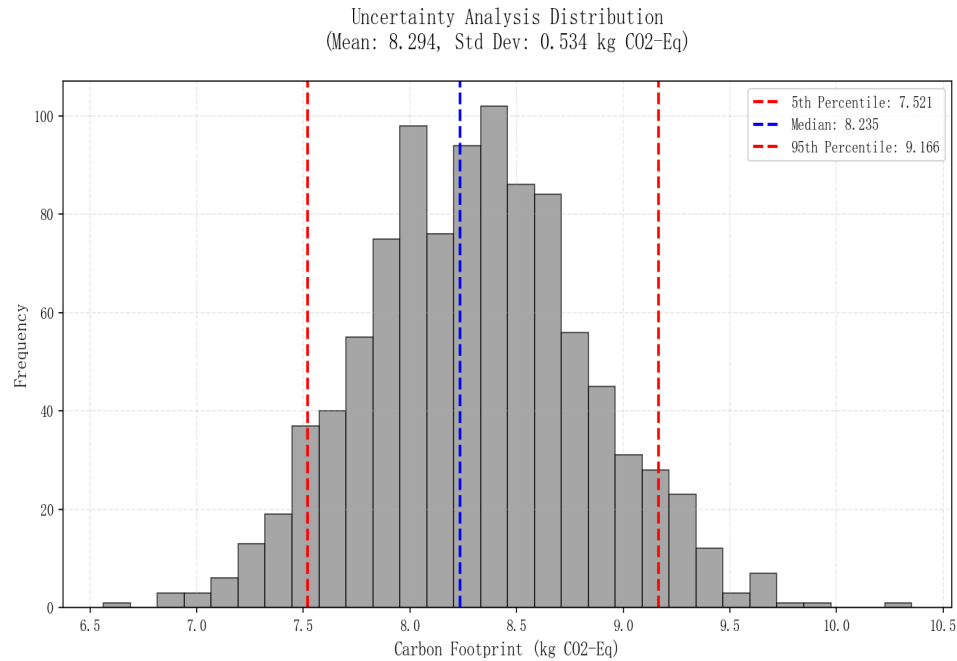
Life Cycle Stage	Carbon Footprint (kg CO2-Eq)	Percentage (%)
Raw Material Acquisition	0.094	1.13
Production	0.229	2.76
Distribution	0.129	1.55

Use	7.646	92.26
End of Life	0.190	2.30
Total	8.287	100.00

3. Uncertainty Analysis Results

Monte Carlo Iterations: 1000

Figure 2: Uncertainty Analysis Distribution



Statistical Metric	Value (kg CO ₂ -Eq)
Mean	8.294
Standard Deviation	0.534
Coefficient of Variation (CV)	6.44%
Median	8.235
5th Percentile	7.521
95th Percentile	9.166

VI. Interpretation of Results

1. Description of Results

The product's total carbon footprint of 8.287 kg CO₂-Eq represents a moderate environmental impact for an electronic component, with the vast majority of emissions occurring during the operational phase rather than upstream or downstream processes. Lifecycle stage contributions exhibit a highly skewed distribution, where the use phase accounts for 92.3% of total emissions, while raw material extraction, production, distribution, and end-of-life stages contribute only 1.1%, 2.8%, 1.6%, and 2.3% respectively. This pattern is characteristic of energy-consuming components integrated into desktop systems, where long-term electricity demand far outweighs manufacturing-related burdens.

The primary emission source is unequivocally the use stage at 7.646 kg CO₂-Eq, driven by continuous power draw during the fan's service life within the power supply unit. In contrast, material and production phases remain marginal due to the component's relatively simple construction and low mass. This dominance indicates that operational efficiency and regional electricity grid characteristics are the decisive factors influencing the overall result, rather than supply-chain or disposal activities.

2. Statement of Assumptions and Limitations

Key assumptions in this study include a standardized product lifespan aligned with typical desktop computer usage patterns, an average European or global electricity mix for the use phase, and generic emission factors for component manufacturing. Additional assumptions encompass uniform load profiles during operation and standard transport distances for distribution, which may not reflect all real-world scenarios or regional variations.

Limitations arise from reliance on secondary databases for material and process data, which may introduce uncertainties in specific alloy compositions or proprietary manufacturing techniques. Data quality for the use phase depends heavily on assumed runtime hours and grid emission factors that can vary significantly by location and time, potentially affecting result robustness. Primary data from the producer were not available, limiting precision for production and end-of-life modeling.

System boundary restrictions encompass a cradle-to-grave approach covering raw materials through disposal, yet exclude upstream infrastructure for electricity generation and certain indirect impacts such as packaging materials or maintenance activities. The analysis also omits potential recycling credits beyond basic end-of-life treatment and does not extend to broader environmental indicators beyond carbon footprint.

3. Suggestions for Improvement

- Prioritize redesign for higher energy efficiency (e.g., low-power brushless motors or optimized blade geometry) to directly reduce the dominant use-phase emissions.
- Source electricity from renewable energy certificates or on-site renewables during manufacturing and encourage end-users to pair the product with green power tariffs.

- Implement design-for-disassembly features to improve end-of-life recovery rates and explore closed-loop material recycling for metals and plastics.
- Collect primary data on actual electricity consumption profiles and regional grid mixes through field measurements to refine use-phase calculations.
- Conduct sensitivity analysis on lifespan assumptions and explore modular fan designs that allow easy replacement, extending overall product service life.